enterprise-okf-ai Handbook

Zero-to-mastery manual for understanding, running, validating, and releasing this repository.

1. Project Purpose

`enterprise-okf-ai` is an enterprise knowledge engineering system that standardizes heterogeneous documentation into an OKF-style bundle and serves grounded retrieval + agentic Q&A.

What this project solves:

- frag

Primary business outcome:

- port

2. Glossary and Definitions

- OKF-style bundle: Directory of markdown concept files with YAML frontmatter metadata and links.

- Fron

3. Prerequisites

- OS: Linux/macOS/Windows (Linux used in this repository run evidence)

- Pyth

4. Environment Setup

4.1 Install dependencies

```
uv sync --extra dev --frozen
```

4.2 Environment variables

Copy and edit:

```
cp .env.example .env
```

Default key settings from `.env.example`:

- `PR

4.3 Config files

- `configs/app.yaml`: base local runtime defaults

- `con

5. Dependency Explanations

Main runtime dependencies from `pyproject.toml` and why they exist:

- `fastapi`, `uvicorn`: API service runtime

- `typ

Quality and release tooling:

- `ruff

6. Repository Structure

High-level map:

.

??? s

Note: `src/okfhub/` exists as a compatibility layer. New implementation focus is `src/enterprise_okf_ai/`.

7. Code Walkthrough

7.1 Ingestion and normalization

- Entry: `enterprise_okf_ai.ingestion.IngestionService`

- Pars

7.2 OKF bundle generation

- Entry: `enterprise_okf_ai.okf.OKFBundleGenerator`
- Res

Required frontmatter keys:

- `id`,

7.3 Validation

- Entry: `enterprise_okf_ai.validators.BundleValidator`
- Cor

7.4 Knowledge graph

- Entry: `enterprise_okf_ai.graph.GraphService`
- Bac

7.5 Embedding and vector indexing

- Entry: `vector_db.indexer.OKFVectorIndexer`
- Stor

7.6 Retrieval

- Entry: `enterprise_okf_ai.retrieval.RetrievalService`
- Cor

Each hit returns score and explanation traces.

7.7 Agent orchestration

- Entry: `enterprise_okf_ai.agent.AgentOrchestrator`

- Cor

Guardrails include evidence thresholds and unsupported-answer handling.

7.8 API and UI

- FastAPI app: `enterprise_okf_ai.api.create_app`

- Stre

8. Training / Inference / Pipeline Flow

This repository does **not** train ML models. It performs retrieval and agent inference over structured knowledge artifacts.

Pipeline flow:

1. Ing

9. Commands Used (Real)

Commands executed in this repository during verification:

```
UV_CACHE_DIR=.uv-cache make check
```

```
UV_CA
```

Observed outputs (latest run):

- lint/t

Source of truth artifact:

- `arti

10. Configuration Explanation

10.1 Runtime settings class

`enterprise_okf_ai.core.settings.Settings` loads `.env` values and resolves relative paths against project root.

Important fields:

- `okf

10.2 Config file usage pattern

- `configs/app.yaml` is a standard environment profile.

- `con
nd ag

11. Validation, Evaluation, and Metrics

11.1 Validation metrics

From latest strict run:

- bun

11.2 Retrieval and graph metrics

From latest strict run:

- grap

11.3 Agent evaluation metrics

From latest strict run (`agent_evaluation.summary`):

- `tota

Interpretation:

- Con

12. Outputs and Interpretation

Primary output directories (gitignored):

- `arti

What to inspect first:

1. `e2

13. Debugging and Troubleshooting

13.1 CLI command not found (`enterprise-okf-ai`)

Use module execution if scripts are not installed in shell path:

```
PYTHONPATH=src uv run --no-sync python -m enterprise_okf_ai.cli.main --help
```

13.2 Git repository appears invalid

If `git` reports not a repository despite `.git` directory presence, initialize a real git repo:

```
git init
git a
```

13.3 GitHub auth invalid

If `gh auth status` reports invalid token:

```
gh auth login -h github.com
```

13.4 Chroma warning in tests

A deprecation warning from `chromadb` may appear (`asyncio.iscoroutinefunction`). It is non-fatal in current runs.

14. Deployment and Execution Notes

Local service startup:

```
make run-api
make
```

FastAPI endpoints to verify:

- `PO

CI workflows:

- `.git

15. Best Practices and Lessons Learned

- Keep bundle generation deterministic to preserve clean git diffs.

- Trea

16. Release Workflow (Recommended)

1. Run:

```
make check
```

```
make
```

2. Prepare release notes (`RELEASE_NOTES.md`) from real metrics.

3. En

17. Official References and Sources

Project references:

- Goo

Official documentation for stack used in this repository:

- uv:

Repository evidence files:

- `arti